Jiangxi University of Finance and Economics

ArchiMate Layered Architecture Report

Luckin Coffee Inc.

A Technology-Driven New Retail Coffee Enterprise

IK2015: Enterprise Architecture and Database Programming

Seminar 6

Group 4

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1 Report Basic Information

Table 1: Basic information of the Seminar 6 report.

Item	Information
Selected enterprise / business	Luckin Coffee Inc.; mobile-first coffee retail, pickup, delivery, partnership-store operation, and packaged coffee business.
Course and seminar	IK2015 Enterprise Architecture and Database Programming; Seminar 6.
Group number	Group 4.
Group members	Zhe-Kai Yang (Krust), Shi-Hang Chen (Sean), Wei-Yu Cai (Nosta), Huan-Wen Chen (Evan), Yuan-Yong Liao (Foreve).
Main deliverable	ArchiMate layered architecture diagrams (Business, Application, Technology) for Luckin Coffee.
Connects to	Seminar 6 requirement: create an ArchiMate-based report for the selected business case study.

This report follows the Seminar 6 instruction sheet. It presents three ArchiMate layered architecture views — Business Layer Model (BLM), Application Layer Model (ALM), and Technology Layer Model (TLM) — that describe the Luckin Coffee enterprise architecture from stakeholder-facing business processes down to a proposed technology deployment view.

2 ArchiMate Layered Architecture Models

Enterprise architecture describes an organization through multiple layers of abstraction. Following the ArchiMate modelling standard, this section presents the Luckin Coffee architecture across three layers: the Business Layer (actors, services, processes), the Application Layer (application services, components, data objects), and the Technology Layer (devices, networks, platform software, and infrastructure artifacts). The Business and Application layers are modelled from the observable Luckin business model and digital service pattern. The Technology Layer is a proposed / assumed deployment architecture for Luckin Coffee based on that digital retail business model; it should not be read as a confirmed disclosure of Luckin Coffee's internal production infrastructure.

2.1 Business Layer Model (BLM)

The Business Layer describes the enterprise from the perspective of business actors, roles, services, processes, and objects. It answers the question: *what does the business do, who is involved, and how is value delivered?*

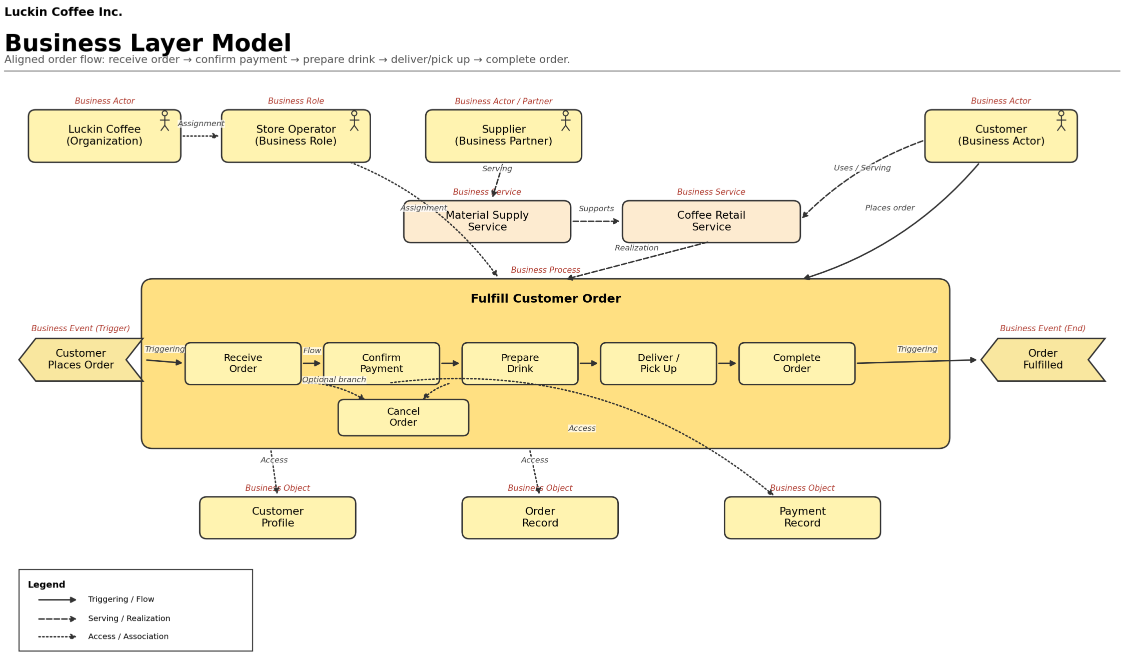


Figure 1: Business Layer Model for Luckin Coffee — showing business actors, roles, the core coffee retail service, and the Fulfill Customer Order business process.

2.1.1 Business Actors and Roles

Table 2: Business actors and roles in the BLM.

Element	ArchiMate Type	Description
Luckin Coffee	Organization	The enterprise itself — the legal entity that owns the brand, platform, and operating model.
Store Operator	Business Role	The operational role responsible for managing store resources, staff scheduling, and physical fulfilment.
Supplier	Business Actor / Partner	External partner providing raw materials (coffee beans, milk, syrup, packaging), equipment, and recipe input.
Customer	Business Actor	The end consumer who places orders, receives beverages, and provides feedback.

2.1.2 Business Service

The model identifies one core business service: **Coffee Retail Service**. This service encapsulates the value Luckin delivers — freshly brewed coffee and tea beverages, fulfilled through pickup or delivery channels, at affordable prices. The Supplier is assigned to support this service, reflecting the integral role of raw-material sourcing in product quality.

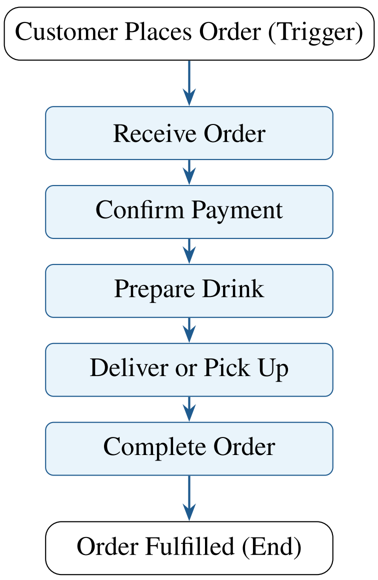
2.1.3 Business Process: Fulfill Customer Order

The central business process is **Fulfill Customer Order**, which decomposes into five sub-processes. In Luckin's usual digital ordering scenario, payment confirmation happens before store preparation and pickup or delivery:

1. **Receive Order** — The customer initiates an order via the mobile app, WeChat mini-program, or other digital channel. The order is captured, validated, and routed to the appropriate store.
2. **Confirm Payment** — The system confirms that the digital payment has succeeded or that an approved payment method has been authorized before fulfilment starts.
3. **Prepare Drink** — Store staff follow standardized recipes to prepare the ordered beverage. Equipment sensors (IoT) may monitor preparation parameters.
4. **Deliver or Pick Up** — The completed order is either handed to the customer for in-store pickup or handed to a delivery rider for last-mile delivery.
5. **Complete Order** — The fulfilment result is recorded after successful pickup or delivery, and the order status is closed.

The process is triggered by the business event **Customer Places Order** and terminates with the end event **Order Fulfilled**. The business object **Customer Profile** is accessed by the Customer actor, linking identity and preference data to the service.

2.1.4 Process Flow Diagram



2.2 Application Layer Model (ALM)

The Application Layer describes the software services, application components, and data objects that realize the business processes. It answers the question: *what applications support the business, and how do they manage data?*

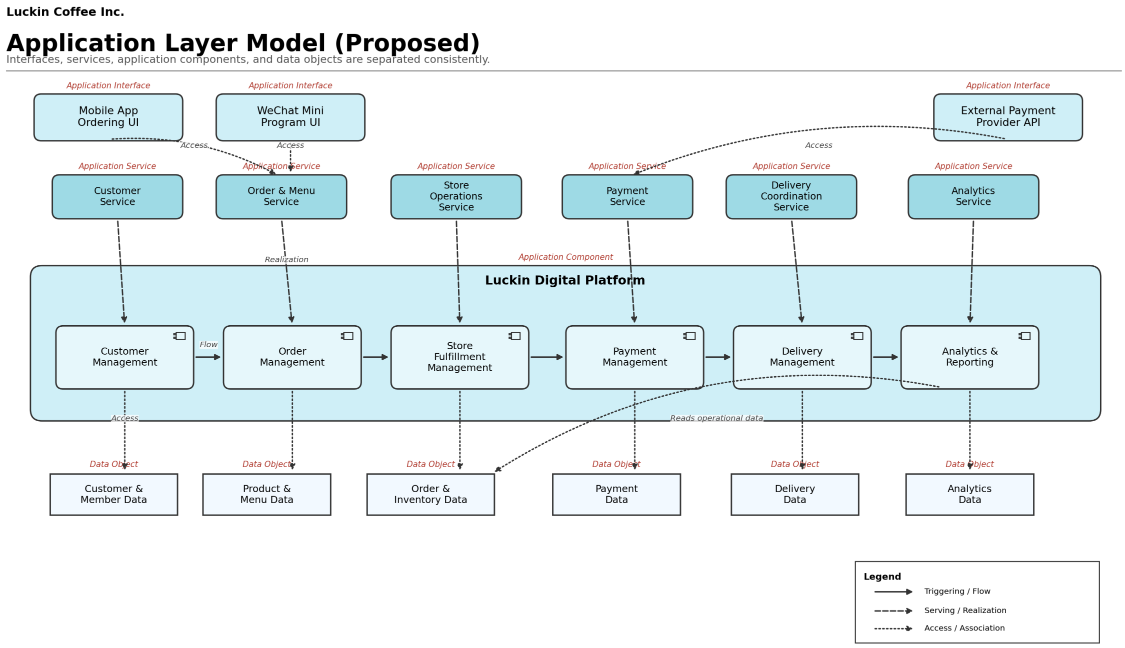


Figure 2: Application Layer Model for Luckin Coffee — showing customer-facing application interfaces, application services, a logical Luckin Digital Platform decomposition, and managed data objects.

2.2.1 Application Interfaces

In ArchiMate, an **Application Interface** is the mechanism through which a user or external system accesses an Application Service. The ALM identifies two primary customer-facing application interfaces:

Table 3: Application interfaces in the ALM.

Interface	Type	Description
Mobile App	iOS / Android Application	Primary customer-facing interface for browsing the menu, placing orders, managing membership, receiving promotions, and tracking order status.
WeChat Mini Program	Mini Program	Lightweight alternative ordering interface embedded in the WeChat ecosystem, reducing the friction of app installation.

2.2.2 Application Services

The ALM exposes four backend application services, each realizing a distinct business capability:

Table 4: Application services in the ALM.

Service	Type	Description
Order & Menu Service	Application Service	Manages the product catalogue (menu), order creation, order routing to stores, and order status lifecycle.
Store Operations Service	Application Service	Supports store-level functions: queue management, preparation workflow, inventory checks, and staff task assignment.
Delivery Coordination Service	Application Service	Coordinates last-mile delivery: rider assignment, route optimization, estimated delivery time, and delivery status tracking.
Payment Service	Application Service	Processes digital payments across multiple methods (WeChat Pay, Alipay, bank card), records transactions, and handles refunds.

2.2.3 Application Components: Luckin Digital Platform

For modelling purposes, the platform is decomposed into six logical application components, each encapsulating a cohesive set of business logic:

Table 5: Application components within the Luckin Digital Platform.

Component	Responsibilities
Customer Management	Customer registration, profile management, membership tiering, points accounting, coupon wallet, and preference tracking.
Order Management	Order creation and validation, order assignment to stores, status tracking, queue management, and order lifecycle from placement to completion.
Store Fulfillment Management	Store-level preparation workflow, recipe display, equipment integration, task queue, and completion confirmation.
Delivery Management	Rider assignment, delivery route optimization, real-time tracking, estimated time of arrival, and delivery completion recording.
Payment Management	Payment processing, transaction logging, refund handling, settlement with payment providers, and payment risk-control support.
Analytics & Reporting	Business intelligence dashboards, sales reports, customer behaviour analysis, campaign effectiveness measurement, and operational KPI computation.

The flow relationships connect application interfaces and services to the components they invoke: the Mobile App and WeChat Mini Program interfaces both serve as access points to the Order & Menu Service, which flows to Order Management; Store Operations Service flows to Store Fulfillment Management; Delivery Coordination Service flows to Delivery Management; and Payment Service flows to Payment Management.

2.2.4 Data Objects

Each application component manages one or more data objects. In the ArchiMate view, these data objects show what information is created, used, or maintained by the platform:

Table 6: Data objects managed by application components.

Data Object	Information Scope	Managed by
Customer & Member Data	Customer identity, membership status, points, coupons, and preferences.	Customer Management
Product & Menu Data	Product catalogue, prices, recipes, availability, and menu presentation.	Order Management
Order & Inventory Data	Customer orders, order items, store stock level, and fulfilment status.	Order & Store Management
Delivery Data	Delivery task, rider assignment, route status, and completion result.	Delivery Management
Payment Data	Payment method, payment confirmation, refund state, and transaction result.	Payment Management
Analytics Data	Sales trend, customer behaviour, promotion performance, and operation indicators.	Analytics & Reporting

2.3 Technology Layer Model (TLM)

The Technology Layer describes a plausible physical and platform infrastructure — devices, networks, platform hosting, system software, and technology artifacts — on which the application components could run. Because Luckin Coffee does not publicly disclose every detail of its internal production infrastructure, this view is treated as a proposed / assumed deployment architecture rather than a verified inventory of actual systems.

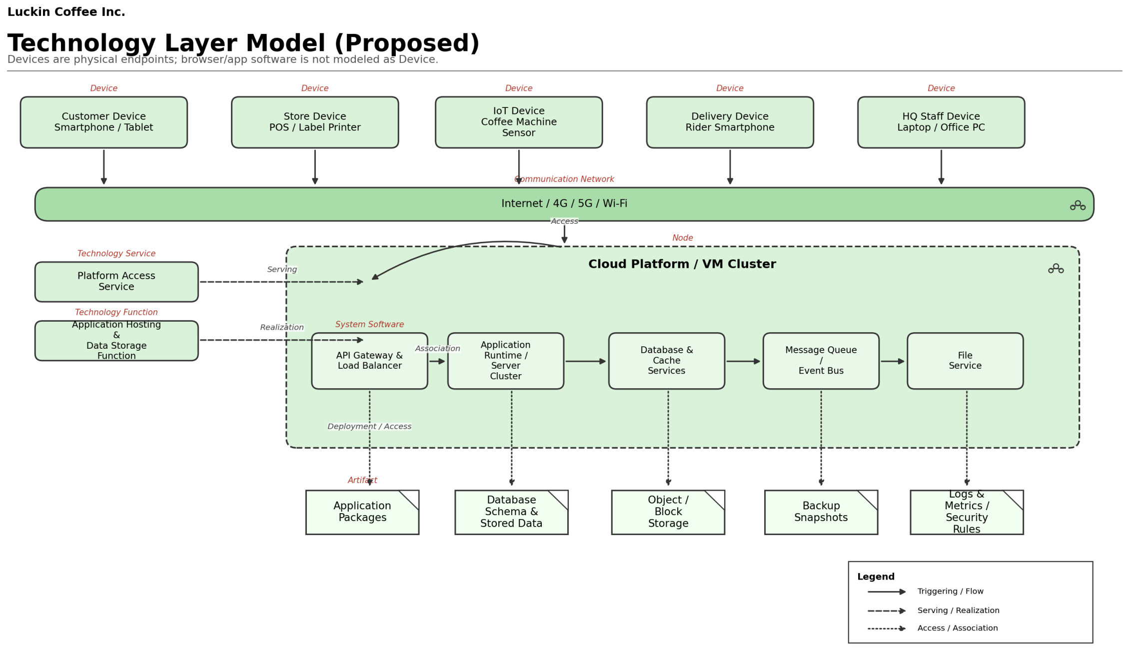


Figure 3: Proposed Technology Layer Model for Luckin Coffee — showing devices, communication networks, a cloud platform, system software, and technology artifacts in an assumed deployment architecture.

2.3.1 Device Layer

Five categories of endpoint or infrastructure devices interact with the Luckin platform. Software entry points such as browsers and rider apps are represented as interfaces running on physical devices, rather than as devices themselves:

Table 7: Device types in the TLM.

Device	Type	Primary Users / Context
Client Device	Smartphone / Tablet	Customers for ordering; Store operators for fulfilment tasks.
Office PC / Laptop	Web Browser Interface	Internal staff (HQ planners, analysts, supply chain managers) access back-office functions through a browser running on a workstation.
Store Device	POS, Label Printer	Store staff for order receipt printing, label generation, and in-store point-of-sale operations.
IoT Device	Coffee Machine Sensors	Store equipment; sensors monitor brewing parameters, machine status, and maintenance needs.
Rider Smartphone	Rider App Interface	Delivery riders use a smartphone-based application interface for receiving tasks, route navigation, and delivery confirmation.

2.3.2 Communication Network

All devices connect through a shared communication layer: **Internet / 4G / 5G / Wi-Fi**. This reflects the practical requirement that Luckin's digital retail architecture must support diverse connectivity — customers on mobile networks, stores on broadband or Wi-Fi, IoT sensors on local wireless, and delivery riders on 4G/5G.

2.3.3 Technology Platform and System Software

The proposed core hosting environment is a **Cloud Platform**. External traffic reaches the platform through a **Platform Access Interface** that routes requests from the communication network into the proposed cloud platform. The following system software elements describe suitable capabilities for this deployment view; product names and technologies are examples, not confirmed Luckin internal implementation details:

Table 8: System software components.

System Software	Function
Application Hosting & Orchestration	Container or virtual-machine based service hosting, deployment, scaling, and health monitoring; Kubernetes is one possible implementation choice.
API Gateway (Load Balancer)	Request routing, rate limiting, authentication, and load distribution across application servers.
Application Servers (Business Logic)	Runtime environment for the six application components of the Luckin Digital Platform.
Data Services (DBMS / Cache / Files)	Relational database management, optional in-memory caching, and file/blob storage; Redis-like caching is one possible implementation choice.
Message Queue (Async Event Bus)	Asynchronous messaging for order events, payment callbacks, delivery status updates, and analytics ingestion.
File Services	Static asset serving (product images, promotional banners), log archival, and report file generation.

2.3.4 Technology Artifacts

The concrete infrastructure artifacts in the proposed cloud deployment are:

Table 9: Technology artifacts.

Artifact	Technology	Purpose
Compute	Containers / VMs	Runs the application servers and hosting layer; can scale with demand during peak morning/lunch hours.
Database	Relational DB / Optional NoSQL Store	Transactional data such as orders, customers, and inventory should be stored in a relational database; semi-structured analytics or log data could use a NoSQL or file-based store.
Storage	Object / Block	Object storage for images, files, backups; block storage for database volumes.
Backup & DR	Backup / Replication Strategy	Disaster recovery should include backup, replication, and point-in-time recovery mechanisms appropriate to the deployment scale.
Monitoring & Security	Logs / Metrics / Security Gateway	Centralized logging, performance metrics dashboards, alerting, and gateway-level security controls such as a WAF where required.

3 Conclusion

This report completed the Seminar 6 task for the Luckin Coffee case study by presenting an ArchiMate-based layered architecture model.

The Business Layer Model (BLM) identified the core business actors (Customer, Store Operator, Supplier), the Coffee Retail Service, and the Fulfill Customer Order process. The Application Layer Model (ALM) mapped this business process to customer-facing interfaces, application services, a logical Luckin Digital Platform decomposition, and managed data objects. The Technology Layer Model (TLM) then described a proposed infrastructure view, including devices, communication networks, cloud platform capabilities, system software, and technology artifacts.

Together, the three layers show how Luckin Coffee's digital retail service can be understood from business operation to application support and technology deployment. The Technology Layer remains a proposed architecture view because Luckin Coffee's internal production infrastructure is not fully public.

References

[1] The Open Group. (2022). *ArchiMate 3.2 Specification*. Van Haren Publishing.

[2] The Open Group. (2022). *TOGAF Standard, 10th Edition*. Van Haren Publishing.

[3] Luckin Coffee Inc. (2025). *FY2025 Annual Report (Form 20-F)*. Available at: https://investor.luckincoffee.com.